DATA VISUALIZATION

Course Code: CSD 601 **Credit Units: 03**

Total Hours: 30

Course Objective:

The course will introduce students to the fundamental of data Visualizations. An understanding of the key techniques and theory used in visualization, including data models, graphical perception and techniques for visual encoding and interaction.

Course Contents:

Module I: Defining Plots: (8 Hours)

bar, line, and stacked charts, axis lengths and limits , plot line styles, properties, and format , Colour Background, colour Setting ticks, labels, and grids histograms , bar plot ,pie chart, scattered plot, using matplotlib and ggplot2.

Module II: Advance Plots: (6 Hours)

Subplots, Data Table and Customizations, transparency and size of axis labels, adding aesthetics, grids, contour plots, Polar plots, colour theory, using matplotlib and ggplot2.

Module III: Plotting With Maps and Images: (8 Hours)

Probability Processing images with PIL, plotting images, Base map, leaflet.js, Captcha Images using matplotlib and ggplot2.

Module IV: Choosing Right Plots: (8 Hours)

Unconstrained optimization; spectrograms, stem plot, streamlines of vector flow, colormaps, scatter plots and histograms, cross-correlation between two variables, Importance of autocorrelation, Latex, Power BI, Tableau.

Course Outcomes:

Upon successful completion of this unit, students should be able to:

- Plot data with right metrics;
- Selection of optimal plots;
- Demonstrate enhanced analytics report,

Examination Scheme:

Components	A	CT	S/V/Q/HA	ESE
Weightage (%)	5	15	10	70

A: Attendance, CT: Class Test, S/V/Q/HA: Seminar/Viva/Quiz/ Home Assignment, EE: End Semester Examination;

Text book & References:

Text books:

- Cole Nussbaumer Knaflic (2015). Storytelling with Data: A Data Visualization Guide for Business Professionals
- Scott Berinato (2016). Goodcharts

Reference Books:

- Robert Collins (2018). Introduction to data visualization with python R, Table

DATA VISUALIZATION LAB

Course Code: CSD 621 **Credit Unit: 01**

Total Hours : 20

Course Objective:

The course will introduce students to the fundamental of data Visualizations. An understanding of the key techniques and theory used in visualization, including data models, graphical perception and techniques for visual encoding and interaction.

Software Requirements: Python 3.6, Anaconda IDE with Spider

Consider this given Data-set for which we will be plotting different charts for experiments 1- 5:

S. No.	Empid	Gender	Age	Sales	BMI	Income
1	E1	M	32	123	NORMAL	350
2	E2	M	35	114	OVERWEIGHT	450
3	E3	F	34	135	OBESITY	170
4	E4	F	42	139	UNDERWEIGHT	190
5	E5	F	26	117	NORMAL	90
6	E6	M	38	121	OBESITY	140
7	E7	F	44	133	OVERWEIGHT	125
8	E8	M	24	140	UNDERWEIGHT	95
9	E9	M	36	133	OVERWEIGHT	200
10	E10	F	40	133	OBESITY	310

List of experiments/demonstrations:

1. Write python code to implement a histogram from above code.
2. Write python code to implement a Column chart from above code
3. Write python code to implement a Box Plot Chart from above code
4. Write python code to implement a Pie Chart from above code
5. Write python code to implement a Scatter Plot from above code
6. Using pandas module write a data frame to csv file
7. Write python code to implement Mandelbrot Fractal Set visualization in Python
8. Write python code to plot Fibonacci spiral fractal using Turtle
9. Implement KDE (Kernel Density Estimate) Plot Visualization with Pandas and Seaborn
10. Implement Violin Plot for Data Analysis.

Course Outcomes:

Upon successful completion of this unit, students should be able to:

- Plot data with right metrics;
- Selection of optimal plots;
- Demonstrate enhanced analytics report,

Examination Scheme:

IA			EE			
A	PR	Practical Based Test	Major Experiment	Minor Experiment	LR	Viva
5	10	15	35	15	10	10

Note: IA –Internal Assessment, EE- External Exam, A- Attendance PR- Performance, LR – Lab Record, V – Viva.

Text Books:

- Cole Nussbaumer Knaflic (2015). Storytelling with Data: A Data Visualization Guide for Business Professionals
- Scott Berinato (2016). Goodcharts

Reference Books:

- Robert Collins (2018). Introduction to data visualization with python R, Table